Accelerator ML Living Review

Summary Statistics

per_year: 12
per_category: 15
per_venue/journal: 53
per_keyword: 17
monthly_trends: 76

Papers

Machine learning as a service system for particle accelerator and its application in CSNS
Mei, Hao, Zhang, Yuliang, Peng, Na, Cheng, Sinong, He, Yongcheng, Xue, Kangjia, Wang, Lin, Li, Mingtao, Wu, Xuan, Zhu, Peng (2025)
Radiation Detection Technology and Methods

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry
Asif Iqbal, John Verboncoeur, Peng Zhang (2025)
arXiv

Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls
Madysa, Penelope, Appel, Sabrina, Kain, Verena, Schenk, Michael (2025)
SoftwareX

Acceleration of Multi-Scale LTS Magnet Simulations with Neural Network Surrogate Models
Louis Denis, Julien Dular, Vincent Nuttens, Mariusz Wozniak, Benoît Vanderheyden, Christophe Geuzaine (2025)
arXiv

Application Of Large Language Models For The Extraction Of Information From Particle Accelerator Technical Documentation
Qing Dai, Rasmus Ischebeck, Maruisz Sapinski, Adam Grycner (2025)
arXiv

Machine Learning approach to classifying quench antenna signals
Plebani, Alberto, Barzi, Emanuela, Teyber, Reed (2025)
Thesis

Machine Learning for superconducting magnets application
Stabilini, Elisa (2025)
Thesis

Batch spacing optimization by reinforcement learning
Remta, Matthias, Velotti, Francesco, Rezagholi, Sharwin (2025)
Phys.Rev.Accel.Beams

Machine learning application for particle accelerator optimization-a review
Isti Dian Rachmawati, Nazrul Effendy, Taufik Taufik (2025)
IAES International Journal of Artificial Intelligence

Machine Learning for Particle Accelerators
Elena Fol, Auralee Edelen (2025)
Unknown Venue

Machine learning based parametrization of the resolution function for the first experimental area of the n_TOF facility at CERN
Petar Žugec, Marta Sabaté-Gilarte, Michael Bacak, Vasilis Vlachoudis, Adria Casanovas, Francisco García-Infantes (2025)
Nuclear Science and Techniques

Integration of Machine Learning-Based Plasma Acceleration Simulations into Geant4: A Case Study with the PALLAS Experiment
A. Sytov, K. Cassou, V. Kubytskyi, M. Lenivenko, A. Huber (2025)
arXiv

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach
Anwar Ibrahim, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin (2025)
arXiv

Physics-Informed Super-Resolution Diffusion for 6D Phase Space Diagnostics
Alexander Scheinker (2025)
arXiv

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations
Hwang, Robin (2024)
arXiv

Virtual Pulse Reconstruction Diagnostic for Single-Shot Measurement of Free Electron Laser Radiation Power
Till Korten, Vladimir Rybnikov, Peter Steinbach, Najmeh Mirian (2024)
Phys.Rev.Accel.Beams

Harnessing Machine Learning for Single-Shot Measurement of Free Electron Laser Pulse Power
Korten, Till, Rybnikov, Vladimir, Vogt, Mathias, Roensch-Schulenburg, Juliane, Steinbach, Peter, Mirian, Najmeh (2024)
Journal

Data-driven gradient optimization for field emission management in a superconducting radio-frequency linac
Goldenberg, Steven, Ahammed, Kawser, Carpenter, Adam, Li, Jiang, Suleiman, Riad, Tennant, Chris (2024)
Phys.Rev.Accel.Beams

Data-Driven Discovery of Beam Centroid Dynamics
Pocher, Liam A., Haber, Irving, Antonsen, Thomas M., O'Shea, Patrick G. (2024)
arXiv

Design and development of advanced Al-Ti-V alloys for beampipe applications in particle accelerators
Kamaljeet Singh, Kangkan Goswami, Raghunath Sahoo, Sumanta Samal (2025)
arXiv

Machine Learning for Reducing Noise in RF Control Signals at Industrial Accelerators
M. Henderson, J. P. Edelen, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2024)
JINST

Beamline Steering Using Deep Learning Models
Allen, Dexter, Kante, Isaac, Bohler, Dorian (2024)
arXiv

Beam-based Identification of Magnetic Field Errors in a Synchrotron using Deep Lie Map Networks
Conrad Caliari, Adrian Oeftiger, Oliver Boine-Frankenheim (2024)
Phys.Rev.Accel.Beams

Linac_Gen: integrating machine learning and particle-in-cell methods for enhanced beam dynamics at Fermilab
Abhishek Pathak (2024)
arXiv

Automated Anomaly Detection on European XFEL Klystrons
Antonin Sulc, Annika Eichler, Tim Wilksen (2024)
arXiv

Accelerator beam phase space tomography using machine learning to account for variations in beamline components
Andrzej Wolski, Diego Botelho, David Dunning, Amelia E. Pollard (2024)
arXiv

Large Language Models for Human-Machine Collaborative Particle Accelerator Tuning through Natural Language
Jan Kaiser, Annika Eichler, Anne Lauscher (2024)
arXiv

Accelerating Cavity Fault Prediction Using Deep Learning at Jefferson Laboratory
Monibor Rahman, Adam Carpenter, Khan Iftekharuddin, Chris Tennant (2024)
arXiv

Anomaly Detection of Particle Orbit in Accelerator using LSTM Deep Learning Technology
Zhiyuan Chen, Wei Lu, Radhika Bhong, Yimin Hu, Brian Freeman, Adam Carpenter (2024)
arXiv

Machine-learning approach for operating electron beam at KEK e^-/e^+ injector Linac
Gaku Mitsuka, Shinnosuke Kato, Naoko Iida, Takuya Natsui, Masanori Satoh (2024)
arXiv

Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations
Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia (2024)
arXiv

Beyond PID Controllers: PPO with Neuralized PID Policy for Proton Beam Intensity Control in Mu2e
Chenwei Xu, Jerry Yao-Chieh Hu, Aakaash Narayanan, Mattson Thieme, Vladimir Nagaslaev, Mark Austin, Jeremy Arnold, Jose Berlioz, Pierrick Hanlet, Aisha Ibrahim, Dennis Nicklaus, Jovan Mitrevski, Jason Michael St. John, Gauri Pradhan, Andrea Saewert, Kiyomi Seiya, Brian Schupbach, Randy Thurman-Keup, Nhan Tran, Rui Shi, Seda Ogrenci, Alexis Maya-Isabelle Shuping, Kyle Hazelwood, Han Liu (2023)
arXiv

Robust Errant Beam Prognostics with Conditional Modeling for Particle Accelerators
Kishansingh Rajput, Malachi Schram, Willem Blokland, Yasir Alanazi, Pradeep Ramuhalli, Alexander Zhukov, Charles Peters, Ricardo Vilalta (2024)
arXiv

Machine Learning For Beamline Steering
Isaac Kante (2023)
arXiv

Variational Autoencoders for Noise Reduction in Industrial LLRF Systems
J. P. Edelen, M. J. Henderson, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2023)
arXiv

Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source
Ryan Humble, William Colocho, Finn O'Shea, Daniel Ratner, Eric Darve (2023)
arXiv

Time-drift Aware RF Optimization with Machine Learning Techniques
R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2023)
arXiv

Distance Preserving Machine Learning for Uncertainty Aware Accelerator Capacitance Predictions
Steven Goldenberg, Malachi Schram, Kishansingh Rajput, Thomas Britton, Chris Pappas, Dan Lu, Jared Walden, Majdi I. Radaideh, Sarah Cousineau, Sudarshan Harave (2023)
arXiv

Learning to Do or Learning While Doing: Reinforcement Learning and Bayesian Optimisation for Online Continuous Tuning
Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia, Oliver Stein, Erik Bründermann, Willi Kuropka, Hannes Dinter, Frank Mayet, Thomas Vinatier, Florian Burkart, Holger Schlarb (2023)
arXiv

Forecasting Particle Accelerator Interruptions Using Logistic LASSO Regression
Sichen Li, Jochem Snuverink, Fernando Perez-Cruz, Andreas Adelmann (2023)
arXiv

Learning Electron Bunch Distribution along a FEL Beamline by Normalising Flows
Anna Willmann, Jurjen Couperus Cabada■, Yen-Yu Chang, Richard Pausch, Amin Ghaith, Alexander Debus, Arie Irman, Michael Bussmann, Ulrich Schramm, Nico Hoffmann (2023)
arXiv

Physics-constrained 3D Convolutional Neural Networks for Electrodynamics
Alexander Scheinker, Reeju Pokharel (2023)
arXiv

Identification of Magnetic Field Errors in Synchrotrons based on Deep Lie Map Networks
Conrad Caliari, Adrian Oeftiger, Oliver Boine-Frankenheim (2023)
arXiv

Data-driven Science and Machine Learning Methods in Laser-Plasma Physics
Andreas Döpp, Christoph Eberle, Sunny Howard, Faran Irshad, Jinpu Lin, Matthew Streeter (2023)
arXiv

Applications of Differentiable Physics Simulations in Particle Accelerator Modeling
Ryan Roussel, Auralee Edelen (2022)
arXiv

Prior-mean-assisted Bayesian optimization application on FRIB Front-End tunning
Kilean Hwang, Tomofumi Maruta, Alexander Plastun, Kei Fukushima, Tong Zhang, Qiang Zhao, Peter Ostroumov, Yue Hao (2022)
arXiv

Fault Prognosis in Particle Accelerator Power Electronics Using Ensemble Learning
Majdi I. Radaideh, Chris Pappas, Mark Wezensky, Pradeep Ramuhalli, Sarah Cousineau (2022)
arXiv

Machine learning-based analysis of experimental electron beams and gamma energy distributions
M. Yadav, M. Oruganti, S. Zhang, B. Naranjo, G. Andonian, Y. Zhuang, Ö. Apsimon, C. P. Welsch, J. B. Rosenzweig (2023)
arXiv

Review of Time Series Forecasting Methods and Their Applications to Particle Accelerators
Sichen Li, Andreas Adelmann (2022)
arXiv

Automatic setup of 18 MeV electron beamline using machine learning
Francesco Maria Velotti, Brennan Goddard, Verena Kain, Rebecca Ramjiawan, Giovanni Zevi Della

Porta, Simon Hirlaender (2022)
arXiv

Diagnostics for Linac Optimization With Machine Learning
R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2022)
arXiv

Transverse phase space tomography in the CLARA accelerator test facility using image compression and machine learning
Andrzej Wolski, Mark A. Johnson, Matthew King, Boris L. Militsyn, Peter H. Williams (2022)
arXiv

Using Kernel-Based Statistical Distance to Study the Dynamics of Charged Particle Beams in Particle-Based Simulation Codes
Chad E. Mitchell, Robert D. Ryne, Kilean Hwang (2022)
arXiv

Neural Network Solver for Coherent Synchrotron Radiation Wakefield Calculations in Accelerator-based Charged Particle Beams
Auralee Edelen, Christopher Mayes (2022)
arXiv

Adaptive Machine Learning for Time-Varying Systems: Towards 6D Phase Space Diagnostics of Short Intense Charged Particle Beams
Alexander Scheinker, Spencer Gessner (2022)
arXiv

Differentiable Preisach Modeling for Characterization and Optimization of Accelerator Systems with Hysteresis
R. Roussel, A. Edelen, D. Ratner, K. Dubey, J. P. Gonzalez-Aguilera, Y. K. Kim, N. Kuklev (2022)
arXiv

Mixed Diagnostics for Longitudinal Properties of Electron Bunches in a Free-Electron Laser
J. Zhu, N. M. Lockmann, M. K. Czwalinna, H. Schlarb (2022)
arXiv

A Neural Network Model of a Quasi-Periodic Elliptically Polarizing Undulator in Universal Mode
Ryan Sheppard, Cameron Baribeau, Tor Pedersen, Mark Boland, Drew Bertwistle (2022)
arXiv

Physics-informed neural network method for modelling beam-wall interactions
Kazuhiro Fujita (2022)
arXiv

Anomaly Detection in Particle Accelerators using Autoencoders
Jonathan P. Edelen, Nathan M. Cook (2021)
arXiv

Input Beam Matching and Beam Dynamics Design Optimization of the IsoDAR RFQ using Statistical and Machine Learning Techniques
Daniel Koser, Loyd Waites, Daniel Winklehner, Matthias Frey, Andreas Adelmann, Janet Conrad (2021)
arXiv

Neural Networks for ID Gap Orbit Distortion Compensation in PETRA III
Bianca Veglia, Ilya Agapov, Joachim Keil (2024)
arXiv

Time-Delayed Koopman Network-Based Model Predictive Control for the FRIB RFQ
Jinyu Wan, Shen Zhao, Wei Chang, Yue Hao (2024)
arXiv

Uncertainty Aware ML-based surrogate models for particle accelerators: A Study at the Fermilab Booster Accelerator Complex
Malachi Schram, Kishansingh Rajput, Karthik Somayaji Peng Li, Jason St. John, Himanshu Sharma (2022)
arXiv

Quantifying Uncertainty for Machine Learning Based Diagnostic
Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
arXiv

Uncertainty Quantification for Virtual Diagnostic of Particle Accelerators
Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
arXiv

Adaptive Latent Space Tuning for Non-Stationary Distributions
Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)
arXiv

Adaptive deep learning for time-varying systems with hidden parameters: Predicting changing input beam distributions of compact particle accelerators
Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)
arXiv

Using LSTM recurrent neural networks for monitoring the LHC superconducting magnets
Maciej Wielgosz, Andrzej Skocze■, Matej Mertik (2017)
arXiv

Coincident Learning for Beam-based RF Station Fault Identification Using Phase Information at the SLAC Linac Coherent Light Source
Jia Liang, William Colocho, Franz-Josef Decker, Ryan Humble, Ben Morris, Finn H. O'Shea, David A. Steele, Zhe Zhang, Eric Darve, Daniel Ratner (2025)
arXiv

Using Convolutional Neural Networks to Accelerate 3D Coherent Synchrotron Radiation Computations
Christopher Leon, Petr M. Anisimov, Nikolai Yampolsky, Alexander Scheinker (2025)
arXiv

Explainable physics-based constraints on reinforcement learning for accelerator controls
Colen, Jonathan, Schram, Malachi, Rajput, Kishansingh, Kasparian, Armen (2025)
arXiv

Accelerator system parameter estimation using variational autoencoded latent regression
Mahindra Rautela, Alan Williams, Alexander Scheinker (2024)
arXiv

Long Short-Term Memory Networks for Anomaly Detection in Magnet Power Supplies of Particle Accelerators
Ihar Lobach, Michael Borland (2024)
arXiv

Leveraging Prior Mean Models for Faster Bayesian Optimization of Particle Accelerators
Tobias Boltz, Jose L. Martinez, Connie Xu, Kathryn R. L. Baker, Zihan Zhu, Jenny Morgan, Ryan Roussel, Daniel Ratner, Brahim Mustapha, Auralee L. Edelen (2025)
arXiv

Optimizing Dynamic Aperture Studies with Active Learning
D. Di Croce, M. Giovannozzi, E. Krymova, T. Pieloni, S. Redaelli, M. Seidel, R. Tomás, F. F. Van der Veken (2024)
arXiv

Neural Network Prior Mean for Particle Accelerator Injector Tuning
Connie Xu, Ryan Roussel, Auralee Edelen (2022)

arXiv

Applications of object detection networks at high-power laser systems and experiments
Jinpu Lin, Florian Haberstroh, Stefan Karsch, Andreas Döpp (2022)
arXiv

A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung
S. Di Carlo, G. Bonvicini, N. A. Althubiti, R. Ayad, E. De La Cruz-Burelo, I. Domínguez, B. O. El Bashir, H. Farhat, J. Flanagan, R. Gillard, S. Izaguirre Gamez, K. Kanazawa, K. Kumara, D. Liventsev, P. L. M. Podesta-Lerma, D. Ricalde-Herrmann, D. Rodriguez Perez, G. Tejeda-Muñoz, M. Tobiyama I. Heredia de la Cruz (2022)
arXiv

Optimizing a Superconducting Radiofrequency Gun Using Deep Reinforcement Learning
David Meier, Luis Vera Ramirez, Jens Völker, Bernhard Sick, Jens Viefhaus, Gregor Hartmann (2022)
arXiv

Uncertainty aware anomaly detection to predict errant beam pulses in the SNS accelerator
Willem Blokland, Pradeep Ramuhalli, Charles Peters, Yigit Yucesan, Alexander Zhukov, Malachi Schram, Kishansingh Rajput, Torri Jeske (2021)
arXiv

Adaptive Machine Learning for Time-Varying Systems: Low Dimensional Latent Space Tuning
Alexander Scheinker (2021)
arXiv

Fast, efficient and flexible particle accelerator optimisation using densely connected and invertible neural networks
Renato Bellotti, Romana Boiger, Andreas Adelmann (2021)
arXiv

Invertible Surrogate Models: Joint surrogate modelling and reconstruction of Laser-Wakefield Acceleration by invertible neural networks
Friedrich Bethke, Richard Pausch, Patrick Stiller, Alexander Debus, Michael Bussmann, Nico Hoffmann (2021)
arXiv

Improving Surrogate Model Accuracy for the LCLS-II Injector Frontend Using Convolutional Neural Networks and Transfer Learning
Lipi Gupta, Auralee Edelen, Nicole Neveu, Aashwin Mishra, Christopher Mayes, Young-Kee Kim (2021)
arXiv

A Novel Approach for Classification and Forecasting of Time Series in Particle Accelerators
Sichen Li, Mélissa Zacharias, Jochem Snuverink, Jaime Coello de Portugal, Fernando Perez-Cruz, Davide Reggiani, Andreas Adelmann (2021)
arXiv

Real-time Artificial Intelligence for Accelerator Control: A Study at the Fermilab Booster
Jason St. John, Christian Herwig, Diana Kafkes, Jovan Mitrevski, William A. Pellico, Gabriel N. Perdue, Andres Quintero-Parra, Brian A. Schupbach, Kiyomi Seiya, Nhan Tran, Malachi Schram, Javier M. Duarte, Yunzhi Huang, Rachael Keller (2021)
arXiv

Surrogate Modeling of the CLIC Final-Focus System using Artificial Neural Networks
J. Ogren, C. Gohil, D. Schulte (2021)
arXiv

Physics-Based Deep Neural Networks for Beam Dynamics in Charged Particle Accelerators
Andrei Ivanov, Ilya Agapov (2020)
arXiv

Introduction to Machine Learning for Accelerator Physics
Daniel Ratner (2020)
arXiv

Machine learning for design optimization of storage ring nonlinear dynamics
Faya Wang, Minghao Song, Auralee Edelen, Xiaobiao Huang (2019)
arXiv

Studies in Applying Machine Learning to LLRF and Resonance Control in Superconducting RF Cavities
Jorge Alberto Diaz Cruz, Sandra Biedron, Manel Martinez-Ramon, Salvador Sosa, Reza Pirayesh (2019)
arXiv

The model of an anomaly detector for HiLumi LHC magnets based on Recurrent Neural Networks and adaptive quantization
Maciej Wielgosz, Matej Mertik, Andrzej Skocze■, Ernesto De Matteis (2017)
arXiv

Machine learning for analysis of plasma driven Ion source
N. Joshi, O. Meusel, H. Podlech (2018)
arXiv

First Steps Toward Incorporating Image Based Diagnostics Into Particle Accelerator Control Systems Using Convolutional Neural Networks
A. L. Edelen, S. G. Biedron, S. V. Milton, J. P. Edelen (2016)
arXiv

Neural Networks for Modeling and Control of Particle Accelerators
A. L. Edelen, S. G. Biedron, B. E. Chase, D. Edstrom, S. V. Milton, P. Stabile (2016)
arXiv

Harnessing the power of gradient-based simulations for multi-objective optimization in particle accelerators
Rajput, Kishansingh, Schram, Malachi, Edelen, Auralee, Colen, Jonathan, Kasparian, Armen, Roussel, Ryan, Carpenter, Adam, Zhang, He, Benesch, Jay (2024)
Mach.Learn.Sci.Tech.

BOOSTR: A Dataset for Accelerator Control Systems
Diana Kafkes, Jason St. John (2021)
arXiv

Model-free and Bayesian Ensembling Model-based Deep Reinforcement Learning for Particle Accelerator Control Demonstrated on the FERMI FEL
Simon Hirlaender, Niky Bruchon (2022)
arXiv

Autonomous Control of a Particle Accelerator using Deep Reinforcement Learning
Xiaoying Pang, Sunil Thulasidasan, Larry Rybarcyk (2020)
arXiv

AI-Assisted Transport of Radioactive Ion Beams
Sergio Lopez-Caceres, Daniel Santiago-Gonzalez (2025)
arXiv

Bayesian Optimization Algorithms for Accelerator Physics
Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lubsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez,

Brahim Mustapha, Verena Kain, Weijian Lin, Simone Maria Liuzzo, Jason St. John, Matthew J. V. Streeter, Remi Lehe, Willie Neiswanger (2024)
arXiv

SPIRAL2 Cryomodules Models: a Gateway to Process Control and Machine Learning
Adrien Vassal, Adnan Ghribi, François Millet, François Bonne, Patrick Bonnay, Pierre-Emmanuel Bernaudin (2021)
arXiv

Predicting Beam Transmission Using 2-Dimensional Phase Space Projections Of Hadron Accelerators
Anthony Tran, Yue Hao, Brahim Mustapha, Jose L. Martinex Marin (2022)
arXiv

Machine Learning for Orders of Magnitude Speedup in Multi-Objective Optimization of Particle Accelerator Systems
Auralee Edelen, Nicole Neveu, Yannick Huber, Mattias Frey, Christopher Mayes, Andreas Adelmann (2020)
arXiv

Machine learning assisted non-destructive transverse beam profile imaging
Zhanibek Omarov, Selcuk Haciomeroglu (2021)
arXiv

GAIA: A General AI Assistant for Intelligent Accelerator Operations
Frank Mayet (2024)
arXiv

AI-Enabled Operations at Fermi Complex: Multivariate Time Series Prediction for Outage Prediction and Diagnosis
Jain, Milan, Mutlu, Burcu O., Stam, Caleb, Strube, Jan, Schupbach, Brian A., John, Jason M.St., Pellico, William A. (2025)
arXiv

Superconducting radio-frequency cavity fault classification using machine learning at Jefferson Laboratory
Chris Tennant, Adam Carpenter, Tom Powers, Anna Shabalina Solopova, Lasitha Vidyaratne, Khan Iftekharuddin (2020)
arXiv

Generative Adversarial Networks (GAN) for compact beam source modelling in Monte Carlo simulations
David Sarrut, Nils Krah, Jean-Michel Létang (2019)
arXiv

Machine learning applied to single-shot x-ray diagnostics in an XFEL
A. Sanchez-Gonzalez, P. Micaelli, C. Olivier, T. R. Barillot, M. Ilchen, A. A. Lutman, A. Marinelli, T. Maxwell, A. Achner, M. Agåker, N. Berrah, C. Bostedt, J. Buck, P. H. Bucksbaum, S. Carron Montero, B. Cooper, J. P. Cryan, M. Dong, R. Feifel, L. J. Frasinski, H. Fukuzawa, A. Galler, G. Hartmann, N. Hartmann, W. Helml, A. S. Johnson, A. Knie, A. O. Lindahl, J. Liu, K. Motomura, M. Mucke, C. O'Grady, J-E. Rubensson, E. R. Simpson, R. J. Squibb, C. Såthe, K. Ueda, M. Vacher, D. J. Walke, V. Zhaunerchyk, R. N. Coffee, J. P. Marangos (2016)
arXiv

Action-Attentive Deep Reinforcement Learning for Autonomous Alignment of Beamlines
Siyu Wang, Shengran Dai, Jianhui Jiang, Shuang Wu, Yufei Peng, Junbin Zhang (2024)
arXiv

Inverse Surrogate Model of a Soft X-Ray Spectrometer using Domain Adaptation
Enrico Ahlers, Peter Feuer-Forson, Gregor Hartmann, Rolf Mitzner, Peter Baumgärtel, Jens Viefhaus (2025)
arXiv

Online tuning and light source control using a physics-informed Gaussian process Adi
A. Hanuka, J. Duris, J. Shtalenkova, D. Kennedy, A. Edelen, D. Ratner, X. Huang (2019)
arXiv

A Two-Stage Machine Learning-Aided Approach for Quench Identification at the European XFEL
Boukela, Lynda, Eichler, Annika, Branlard, Julien, Jomhari, Nur Zulaiha (2024)
arXiv

Beam Detection Based on Machine Learning Algorithms
Haoyuan Li, Qing Yin (2023)
arXiv

Accurate and confident prediction of electron beam longitudinal properties using spectral virtual diagnostics
A. Hanuka, C. Emma, T. Maxwell, A. Fisher, B. Jacobson, M. J. Hogan, Z. Huang (2020)
arXiv

A Start To End Machine Learning Approach To Maximize Scientific Throughput From The LCLS-II-HE
Aashwin Mishra, Matt Seaberg, Ryan Roussel, Fred Poitevin, Jana Thayer, Daniel Ratner, Auralee Edelen, Apurva Mehta (2025)
arXiv

A Conceptual Development of Quench Prediction App build on LSTM and ELQA framework
Matej Mertik, Maciej Wielgosz, Andrzej Skocze■ (2016)
arXiv

Symplectic machine learning model for fast simulation of space-charge effects
Wan, Jinyu, Qiang, Ji, Hao, Yue (2025)
Phys.Rev.Accel.Beams

Laser wakefield electron acceleration simulation using physics-informed diffusion probabilistic models
Jech, Matej, Kovalenko, Alexander, Lazzarini, Carlo Maria, Grittani, Gabriele Maria (2025)
Proc.SPIE Int.Soc.Opt.Eng.

Using a neural network model to guide protection heater design in Nb$_{3}$Sn accelerator magnets
Bakrani Balani, Shahriar, Salmi, Tiina (2025)
Supercond.Sci.Technol.

Reinforcement Learning for Charged Particle Beam Control to Minimize Injection Mismatch in Particle Accelerators
Balasooriya, Thilina, Yoo, Shinjae, Schoefer, Vincent, Tseng, Huan-Hsin, Gao, Yuan, Lin, Weijian, Silva, Chanaka De (2025)
Journal

Outlook towards deployable continual learning for particle accelerators
Rajput, Kishansingh, Lin, Sen, Edelen, Auralee, Blokland, Willem, Schram, Malachi (2025)
Mach.Learn.Sci.Tech.

Application of ensemble machine learning algorithms and filtering techniques in slow orbit feedback systems of electron storage rings
Fan, Jiaqi, Liu, Weibin, Wang, Jiuqing, Wei, Yanru, Wei, Yuanyuan, Ji, Daheng (2025)
Phys.Rev.Accel.Beams

femto-PIXAR: a self-supervised neural network method for reconstructing femtosecond X-ray free electron laser pulses
Goetzke, Gesa, Plumley, Rajan, Hartmann, Gregor, Maxwell, Tim, Decker, Franz-Josef, Lutman, Alberto, Dunne, Mike, Ratner, Daniel, Turner, Joshua J. (2025)
Opt.Express

Deep Lie Map Networks:A novel Approach to Infer Nonlinear Synchrotron Optics from Beam Oscillations
Caliari, Conrad (2025)
Thesis

Reconstructing time-of-flight detector values of angular streaking using machine learning
Meier, David, Viefhaus, Jens, Hartmann, Gregor, Helml, Wolfram, Otto, Thorsten, Sick, Bernhard (2025)
Phys.Rev.Accel.Beams

Artificial intelligence for advancing particle accelerators
Ghribi, Adnan, Cassou, Kevin, Dalena, Barbara, Eichler, Annika, Guler, Hayg, Mistry, Andrew K., Oeftiger, Adrian, Shea, Thomas, Valentino, Gianluca, Welsch, Carsten P. (2025)
Europhys.News

Machine Learning for Optimized Polarization at Jefferson Lab
Jeske, Torri, Kasparian, Armen, Lawrence, David, Britton, Thomas, Schram, Malachi, Moran, Patrick, Fanelli, Cristiano, Guo, Jiawei, Jarvis, Naomi, Maxwell, James, Keith, Chris (2025)
EPJ Web Conf.

Applying the machine learning methods to determine the linear optics parameters in the ThomX collector ring
Klekots, D., Bezshyyko, O., Golinka-Bezshyyko, L., Kubytskyi, V., Chaikovska, I. (2024)
Nucl.Phys.Atom.Energy

Deep learning framework for fault detection in accelerators
Piekarski, Michal (2024)
JACoW

Machine Learning Applications for Improving Accelerator Operations
Lin, Lucy (2024)
Thesis

Assessing the Performance of Deep Learning Predictions for Dynamic Aperture of a Hadron Circular Particle Accelerator
Di Croce, Davide, Giovannozzi, Massimo, Montanari, Carlo Emilio, Pieloni, Tatiana, Redaelli, Stefano, Van der Veken, Frederik F. (2024)
Instruments

Machine learning
Snuverink, Jochem (2024)
CERN Yellow Rep.School Proc.

Third-integer Resonant Extraction Regulation System for Mu2e
Narayanan, Aakaash (2024)
InspireHEP

Machine learning-based non-destructive measurement of bunch length at FRIB
Wan, Jinyu, Plastun, Alexander, Ostroumov, Peter (2024)
JACoW

Beam emittance and Twiss parameters from pepper-pot images using physically informed neural nets
Knight, Ian, Mustapha, Brahim (2024)
JACoW

Advancements in backwards differentiable beam dynamics simulations for accelerator design, model calibration, and machine learning
Roussel, Ryan, Edelen, Auralee, Gonzalez-Aguilera, Juan Pablo, Lehe, Remi, Huebl, Axel, Kaiser, Jan, Santamaria Garcia, Andrea, Xu, Chenran, Eichler, Annika, Charleux, Grégoire (2024)
JACoW

Advanced algorithms for linear accelerator design and operation
Ong, Ysabella Kassandra, Bellan, Luca, Pisent, Andrea, Comunian, Michele, Fagotti, Enrico, Bortolato, Damiano, Montis, Maurizio, Giacchini, Mauro, Carletto, Osvaldo (2024)
JACoW

Adaptive machine learning with hard physics constraints and generative diffusion for 6D phase space diagnostics
Scheinker, Alexander (2024)
JACoW

Towards Agentic AI on Particle Accelerators
Antonin Sulc, Thorsten Hellert, Raimund Kammering, Hayden Hoschouer, Jason St. John (2025)
Journal

Continuous data-driven control of the GTS-LHC ion source at CERN
Kain, Verena, Rodriguez Mateos, Borja, Bruchon, Niky, Hirlaender, Simon, Küchler, Detlef (2024)
JACoW

Surrogate Models studies for laser-plasma accelerator electron source design through numerical optimisation
G. Kane, P. Drobniak, S. Kazamias, V. Kubytskyi, M. Lenivenko, B. Lucas, J. Serhal, K. Cassou, A. Beck, A. Specka, F. Massimo (2025)
arXiv

Reinforcement learning-trained optimisers and Bayesian optimisation for online particle accelerator tuning
Kaiser, Jan, Xu, Chenran, Eichler, Annika, Santamaria Garcia, Andrea, Stein, Oliver, Bründermann, Erik, Kuropka, Willi, Dinter, Hannes, Mayet, Frank, Vinatier, Thomas, Burkart, Florian, Schlarb, Holger (2024)
Sci.Rep.

Machine learning-based extraction of longitudinal beam parameters in the LHC
Iliakis, Konstantinos, Karlsen-Bæck, Birk Emil, Trad, Georges, Timko, Helga, Zampetakis, Michail, Argyropoulos, Theodoros (2024)
JACoW

High fidelity numerical modelling and condition monitoring applied to septum magnets at CERN
Kawa, Krzysztof, Szumlak, Tomasz, Kokkinos, Charilaos, Lackner, Friedrich (2024)
JACoW

Updates to Xopt for online accelerator optimization and control
Roussel, Ryan, Kennedy, Dylan, Boltz, Tobias, Baker, Kathryn, Mayes, Christopher, Edelen, Auralee (2024)
JACoW

Breaking new ground in data-intensive science: first insights from the LIV.INNO center for doctoral training
Welsch, Carsten (2024)
JACoW

The reinforcement learning for autonomous accelerators collaboration
Santamaria Garcia, Andrea, Eichler, Annika, Xu, Chenran, Kaiser, Jan, Scomparin, Luca, Schenk, Michael, Pochaba, Sabrina, Hirlaender, Simon (2024)
JACoW

Machine learning-based particle accelerator modeling
Emmanuel Goutierre (2024)
Laboratoire Interdisciplinaire des Sciences du Numérique

Modelization of an Injector With Machine Learning
Mathieu Debongnie, Maud Baylac, Frédéric Bouly, Nicolas Chauvin, Angélique Gatera, Tomas Junquera, Didier Uriot (2019)

10th International Particle Accelerator Conference

A Machine Learning Technique for Dynamic Aperture Computation
Barbara Dalena, Mehdi Ben Ghali (2021)
12th International Particle Accelerator Conference

Surrogate Model for Linear Accelerator: A fast Neural Network approximation of ThomX's simulator
Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2023)
IPAC 2023 - 14th International Particle Accelerator Conference

Physics-aware modelling of an accelerated particle cloud
Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2023)
MLPS 2023 - Machine Learning and the Physical Sciences Workshop 23023 - At the 37th conference on Neural Information Processing Systems (NeurIPS)

From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale
Axel Huebl, Remi Lehe, Edoardo Zoni, Olga Shapoval, Ryan T. Sandberg, Marco Garten, Arianna Formenti, Revathi Jambunathan, Prabhat Kumar, Kevin Gott, Andrew Myers, Weiqun Zhang, Ann Almgren, Chad E. Mitchell, Ji Qiang, David Grote, Alexander Sinn, Severin Diederichs, Maxence Thevenet, Luca Fedeli, Thomas Clark, Neil Zaim, Henri Vincenti, Jean-Luc Vay (2023)
20th Advanced Accelerator Concepts Workshop (AAC'22)

RBF neural net based classifier for the AIRIX accelerator fault diagnosis
J. C. Ribes, G. Delaunay, J. Delvaux, E. Merle, M. Mouillet (2000)
arXiv

A Living Review Pipeline for AI/ML Applications in Accelerator Physics
Adnan Ghribi (2025)
arXiv

Digital Twin Framework for PIP-II Linac: AI-Driven Multi-Scale Modeling from Ion Source to 800 MeV
Pathak, Abhishek, Hanlet, Pierrick, Miceli, Tia (2025)
Journal

Reinforcement Learning for Accelerator Beamline Control: a simulation-based approach
Anwar Ibrahim, Alexey Petrenko, Maxim Kaledin, Ehab Suleiman, Fedor Ratnikov, Denis Derkach (2025)
arXiv (Cornell University)

Automated beam tuning of the TRIUMF ISAC facility
Shelbaya, Olivier, Jung, Paul, Kester, Oliver, Hassan, Omar (2025)
JACoW

Beam Energy Forecasting using Machine Learning at the CLEAR accelerator
Gilardi, Antonio, Malyzhenkov, Alexander, Petersson,Alfred, Mostacci Andrea, Pollastro, Andrea, Aksoy, Avni, Filippetto, Daniele, Gamba,Davide, Granados, Eduardo, Tangari,Giacomo, Sjobak, Kyrre, Bonnard,Ladislas, Wroe,Laurence, Carranza-García, Manuel, Franek Ondrej, Korysko,Pierre, Corsini Roberto, Rieker,Vilde, Farabolini, Wilfrid, Mazzoni, Stefano (2025)
JACOW

Advancing accelerator science through data-intensive research and training
Welsch, Carsten (2025)
JACOW

Machine learning approach to MDI optimization for 3 TeV c.o.m. Muon Collider
Castelli, Luca (2025)
JACOW

Advances in machine learning inference of dynamic aperture evaluation for the LHC
Montanari,Carlo Emilio, Di Croce, Davide, Van der Veken,Frederik, Giovannozzi, Massimo, Appleby, Robert, Redaelli, Stefano, Pieloni, Tatiana (2025)
JACOW

Machine learning for calibration drift forecasting in superconducting RF cavities
Sun Yue, Bellandi Andrea, Eichler, Annika, Richter,Bozo, Diomede,Marco, Schmidt, Christian, Schlarb Holger, Branlard, Julien, Herrmann,Max (2025)
JACOW

Uncertainty-Quantified Machine Model Construction Using Physics-Informed Gaussian Processes and Bayesian Optimization
Isensee,Victoria, Oeftiger, Adrian, Boine-Frankenheim, Oliver (2025)
JACOW

ChatEED: An agentic retrieval assistant for accelerator operators
Aaron Zachary Reed, Claudio Bisegni, Sandesh Shrestha, Michelle Huang, Daniel Ratner (2025)
Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis

Automated RF Phase Adjustment for Beam Stabilization in the Fermilab Linac
Chichili, R.R., Sulskis, J.A., Sharankova, R., Vamanan, B., Ravi, S. (2025)
Journal

Fast Adaptive Neural Control of Resonant Extraction at Fermilab
Whitbeck, A., Berlioz, J., Danison-Fieldhouse, K., Hazelwood, K., Khan, M., Mitrevski, J., Narayanan, A., John, J.St., Tran, N., Ji, J., Walter, M. (2025)
Journal

First Experimental Demonstration of Machine Learning-Based Tuning on the PSI Injector 2 Cyclotron
M. Haj Tahar, W. Joho, E. Solodko, M. Bocchio, S. Marquie, M. Busch, A. Barchetti, J. Grillenberger, J. Snuverink, M. Schneider (2025)
arXiv (Cornell University)

Xopt and Badger: a machine learning ecosystem for real-time accelerator control and optimization
Roussel, Ryan, Kennedy,Dylan, Edelen, Auralee, Kuklev, Nikita, Miskovich,Sara, Zhang Zhe (2025)
JACOW

Long short-term memory of recurrent neural network to the analysis of temperature rise on the production target in Hadron Experimental Facility of J-PARC
Agari Keizo, Sato Yoshinori, Toyoda Akihisa, Morino Yuhei, Akiyama Hironobu (2025)
JACOW

Autonomous Pressure Control in MuVacAS via Deep Reinforcement Learning and Deep Learning Surrogate Models
Rodriguez-Llorente, Guillermo, Gallardo, Galo, Navascués, Rodrigo Morant, Petrovsky, Nikita Khvatkin, Sabogal, Anderson, Martín, Roberto Gómez-Espinosa (2025)
Journal

Machine learning-based prediction of magnet errors in storage ring light sources
Xu, Jianhao (2025)
arXiv

Explainable physics-based constraints on reinforcement learning for accelerator optimization
Jonathan Colen, Malachi Schram, Kishansingh Rajput, Armen Kasparian (2026)
Machine Learning: Science and Technology

U-Net-based surrogate modeling for attosecond X-ray free-electron lasers
Yufei Wei, Bingyang Yan, Chenzhi Xu, Jiawei Yan, Haixiao Deng (2026)
arXiv

Towards continual machine learning for particle accelerators
Rajput, Kishansingh, Schram, Malachi, Blokland, Willem, Zhukov, Alexander, Lin, Sen (2026)
JACoW

JuTrack, a Julia-based auto-differentiable accelerator simulation code for advanced dynamics, scientific machine learning and optimization
Wan, Jinyu, Alamprese, Helena, Ratcliff, Christian, Qiang, Ji, Hao, Yue (2026)
JACoW

Data-driven beam diagnostics using betatron radiation
Saberi, Hossein, Burvill, Jasper, Siddle, Carys, Xia, Guoxing, Patrick Farmer, John, Pukhov, Alexander (2026)
Plasma Phys.Control.Fusion

High-Resolution Image Reconstruction with Unsupervised Learning and Noisy Data Applied to Ion-Beam Dynamics for Particle Accelerators
Francis Osswald, Mohammed Chahbaoui, X. Liang (2026)
Journal of Instrumentation

A Hybrid Monte Carlo–Machine Learning Framework for High-Energy Neutron Shielding Using Boron-Enhanced Concrete
Demet Sariyer, Elif Y■ld■r■m (2026)
International Journal of Computational and Experimental Science and Engineering

Hybrid machine learning-monte carlo approach for neutron shielding in high-energy proton facilities
Sariyer, Demet (2026)
J.Subatomic Part.Cosmol.

Machine learning-enhanced infrared imaging for temperature anomaly detection in power supplies
Mohsen, Osama, Lobach, Ihar, Borland, Michael, Sun, Yine (2026)
JACoW

Anomaly detection of slow-moving variables at LANSCE for improved beam quality
Huang, En-Chuan, Yampolsky, Nikolai (2026)
JACoW

Online optimizations of NSLS-II Linac and Linac-to-Booster beam lines using machine learning methods
Song, Minghao, Wang, Guimei, Hidaka, Yoshiteru, Yang, Xi (2026)
JACoW

Machine learning-driven computations of 3D Coherent Synchrotron Radiation
Leon, Christopher, Scheinker, Alexander, Yampolsky, Nikolai, Anisimov, Petr M. (2026)
JACoW

Safe extremum seeking for real-time adaptive accelerator control
Williams, Alan, Scheinker, Alexander (2026)
JACoW

Machine learning assisted Bayesian calibration of accelerator digital twin from orbit response data
Lin, Weijian, Kelly, Christopher, Hoffstaetter, Georg, Brown, Kevin, Urban, Nathan (2026)
JACoW

Integrating simulation and machine learning for Proton Storage Ring beam analysis
Leon, Christopher, Huang, En-Chuan, Scheinker, Alexander, Anisimov, Petr M. (2026)
JACoW

Towards differentiable beam dynamics modeling in BLAST/ImpactX
Huebl, Axel, Mitchell, Chad, Lehe, Remi, Charleux, Grégoire, Myers, Andrew, Zhang, Weiqun, Qiang, Ji, Vay, Jean-Luc, Kaiser, Jan, Hespe, Christian, Gonzalez-Aguilera, Juan Pablo, Xu,

Chenran, Santamaria Garcia, Andrea, Roussel, Ryan, Edelen, Auralee, Moses, William Steven (2026)
JACoW

A self-supervised transformer for RF cavity signal denoising
Rajesh, Vikshar, Edelen, Auralee, Hall, Christopher, O'Shea, Finn, Edelen, Jonathan (2026)
JACoW

Machine learning using beam loss monitors for DIAMOND-II
Lehmann, Corey, Bobb, Lorraine, Burrows, Philip (2026)
JACoW

Beam diagnostics, data acquisition system, and applications of machine learning at the KEK e-/e+ Linac
Miyahara, Fusashi, Kinoshita, Kaito, Kurata, Masakazu, Iwasaki, Masako, Natsui, Takuya, Okayasu, Yuichi, Satake, Itsuka, Satoh, Masanori, Uemura, Kosuke, Wang, Di (2026)
JACoW

Machine Learning for Anomaly Detection in Neural Network Security and SRF Cavities
Hal Ferguson (2026)
ODU Digital Commons (Old Dominion University)

Continual Learning for Particle Accelerators
Nuclear Physics (NP) USDOE Office of Science (SC), M. Schram, Thomas Jefferson National Accelerator Facility (TJNAF), Sen Lin, Kishansingh Rajput, Alexander Zhukov, Willem Blokland (2026)
Unknown Venue

Advancing accelerator virtual beam diagnostics through latent evolution modeling: an integrated solution to forward, inverse, tuning, and UQ problems
Rautela, Mahindra, Scheinker, Alexander (2026)
arXiv

Machine learning at the Spallation Neutron Source accelerator and target
Armen Kasparian, Willem Blokland, Alexander Zhukov, Anant Ray, Drew Winder, Kishansingh Rajput, Malachi Schram (2026)
JACOW

Machine learning-enhanced deterministic controls in lasers and accelerators
Dan Wang, Alessio Amodio, Alex Picksley, A. J. Gonsalves, Curtis J. Berger, Hai-En Tsai, Jeroen van Tilborg, Mahek Logantha, N. R. Vora, Qiang Du, Russell Wilcox, Samuel K. Barber, Tong Zhou, Z. Eisentraut (2026)
JACOW

Method for reconstruction the energy spectrum of a linear electron accelerator using neural networks
A.A. Kim, F.R. Studenikin, V.V. Khankin, S.A. Zolotov, A.P. Lyubomudrov, A.P. Chernyaev (2026)
Radiation Physics and Chemistry

AI-Ready Control System for the Fermilab Accelerator Complex
Tia Miceli, Erik Gottschalk, Donovan Tooke, Evan Milton, Robert Santucci, Hayden Hoschouer, Michael Balcewicz, Jennifer Case, Abhishek Deshpande, Kit Fieldhouse, Sudeshna Ganguly, Beau Harrison, Aisha Ibrahim, Thomas Kobilarcik, Michael Olander, Abhishek Pathak, Jason St. John, Aaron Sauers (2026)
arXiv

Machine learning-based virtual diagnostics of dielectric laser acceleration
Thilo Egenolf, Oliver Boine-Frankenheim (2026)
arXiv (Cornell University)

Data-Driven Optimisation of Superconducting Magnets at CEA Paris-Saclay
Damien F. G. Minenna, Guillaume Dilasser, Robin Penavaire, Valerio Calvelli, Thibault de Chabannes, Thibault Lecrevisse, Thomas Achard, Jason Le Coz, Christophe Berriaud, Benoît

Bolzon, Antomne Caunes, Phillipe Fazilleau, Hélène Felice, Clément Genot, Antoine Guinet, Nikola Jerance, François-Paul Juster, Thibaut Lemercier, Gilles Lenoir, Clément Lorin, Yann Perron, Camille Pucheu-Plante, Étienne Rochepault, Damien Simon, Francesco Stacchi, Michel Segreti, Vincent Trauchessec, Olivier Tuske, Hajar Zgour (2026)
EPJ Research Infrastructures

Microsecond-latency feedback at a particle accelerator by online reinforcement learning on hardware
L. Scomparin, Michele Caselle, Andrea Santamaria Garcia, Chenran Xu, Edmund Blomley, Timo Dritschler, Akira Mochihashi, Marcel Schuh, Johannes Leonhard Steinmann, Erik Bründermann, Andreas Kopmann, Jürgen Becker, A.-S. Mueller, Marc Weber (2026)
Machine Learning Science and Technology

Machine-State Embeddings as an Operational Coordinate System for Accelerator Operation
Chris Tennant, Jundong Li, Song Wang (2026)
arXiv

Adaptive Control Meets Deep Learning: EXTREMUM SEEKING IN LATENT SPACE WITH PARTICLE ACCELERATOR APPLICATIONS
Alexander Scheinker (2026)
IEEE Control Systems

Closing the Loop: Deploying Auto-Generating Digital Twins for Particle Accelerators
A. D. Brynes, M. King, K. R. L. Baker, R. Banerjee, R. Clarke, D. J. Dunning, J. K. Jones, M. Leputa, A. E. Pollard, M. Romanovschi, M. Shaw, N. Ziyan (2026)
arXiv

RL-ABC: Reinforcement Learning for Accelerator Beamline Control
Anwar Ibrahim, Fedor Ratnikov, Maxim Kaledin, Alexey Petrenko, Denis Derkach (2026)
arXiv

Predicting LTS Superconducting NbTi Magnet Quench: A 1D-CNN Model for Real-Time Implementations
Aline Y. Machado, Pedro T. Andrade, Amanda C. Martinez (2026)
IEEE Transactions on Applied Superconductivity

Development of an Experimental Test Stand and Machine Learning Surrogate Models for Capturing Complex Responses in Pulsed Septa
Kawa, Krzysztof, Szumlak, Tomasz, Yagci, Omer Yusuf, Balhan, Bruno, Lackner, Friedrich (2026)
IEEE Trans.Appl.Supercond.

Machine learning for mapping multipactor susceptibility in RF systems: Capabilities and generalization constraints
Asif Iqbal, John Verboncoeur, Peng Zhang (2026)
APL Machine Learning

Rapid Secondary Neutron Dose Estimation in Proton Accelerators Using FLUKA and Machine Learning
Demet Sariyer, Elif Y■ld■r■m (2026)
Süleyman Demirel Üniversitesi Fen Edebiyat Fakültesi Fen Dergisi

Lightweight CNN-Based Anomaly Detection for High Voltage Converter Modulators in the Spallation Neutron Source
Cencillo, Alberto D., Concepción, Leonardo, Luengo, Julián, Triguero, Isaac (2026)
arXiv

Learning Dynamic Aperture from One-turn Maps
Derong Xu (2026)
arXiv

A hybrid monte Carlo-machine learning framework for neutron dose attenuation modeling in layered iron-concrete shields of a 1000 MeV proton accelerator
Demet Sar■yer, Ali Osman GÖKCAN (2026)

Annals of Nuclear Energy

Beam-Level Nonlinear Compton Spectra via a Neural Network Surrogate Model
Antonina Timoshenko, Maxim Malakhov, Alexander Fedotov, Sergey Rykovanov (2026)
arXiv

Machine Learning-Driven Beam Tuning Using Adaptive Region Bayesian Optimization at INFN-LNL
Ysabella Kassandra Ong, Luca Bellan, D. Bortolato, Maurizio Montis, M. Comunian, Natalia Milas, Ryoichi Miyamoto, Domenic Nicosia, Francesco Grespan, E. Fagotti, A. Pisent (2026)
Instruments

Report on the Advanced Linear Collider Study Group (ALEGRO) Workshop 2026
L. Verra, P. Muggli, B. Cross, E. Adli, R. Babjak, T. Barklow, F. Bencivenga, C. Benedetti, C. Benedetti, M. Buscher, S. S. Bulanov, A. Caldwell, G. Chen, O. Chubenko, R. D'Arcy, S. Diederichs, K. Downham, J. Farmer, M. Ferrario, A. Ferran Pousa, A. Formenti, M. Fuchs, S. Gessner, L. Giannessi, Z. Gong, J. Grames, T. Grismayer, H. Jaworska, L. Ji, V. Kain, S. Karkare, A. Knetsch, C. Lindstrom, N. Lopes, N. Lopes, F. Massimo, C. Masciovecchio, P. Oliveira, T. Opferkuch, J. Osterhoff, J. Osterhoff, S. Pagan Griso, G. Penco, J. Power, A. Pukhov, T. P. Rakitzis, A. Rastogi, L. Reichwein, C. B. Schroeder, L. O. Silva, A. Sinn, D. Sofikitis, D. Storey, M. Thévenet, M. Trovò, M. Turner, M. Vranic, F. Willeke, T. Wilson, M. Wing, W. Zhang, Chuan Zheng (2026)
arXiv (Cornell University)

Robust Betatron-Tune Measurement from Schottky Spectra: Complementary Classical and Deep-Learning Paradigms
Peihan Sun, Manzhou Zhang, Renxian Yuan, Deming Li, Jian Dong (2026)
arXiv

Smart prediction of beam availability at synchrotron facilities
Mustafa Alzu’bi, Nailah Al-Madi, Rawan Ghnemat (2026)
Discover Artificial Intelligence

Application of quantum machine learning using variational quantum classifier in accelerator physics
Yin, He-Xing, Hu, Zhi-Yuan, Zeng, Huan-Huan, Guan, Jia-Bao, Wang, Ji-ke (2025)
Nuclear Science and Techniques

A corrective agentic hybrid RAG and an operations-grounded evaluation for a scientific facility
Rajat Sainju, Dariusz Jarosz, Hairong Shang, Michael Prince, Ryan M. Aydelott, Mathew J. Cherukara, Yine Sun, Michael D. Borland (2026)
arXiv

Predict before you train: Scaling Laws for particle physics foundation models
Uslu, Jan-Lucas, Nachman, Benjamin, Re, Christopher (2026)
arXiv

Reinforcement Learning applied to Optimization of LHC beams in the CERN Proton Synchrotron
Joel Axel Wulff, Alexandre Lasheen (2026)
arXiv

Simulated six-dimensional beam-dynamics dataset for machine-learning surrogate modelling
Santiago Martinez Saenz (2026)
Zenodo (CERN European Organization for Nuclear Research)

Deep-Learning-Enhanced Beam Diagnostics and High-Resolution Image Denoising for Next-Generation Particle Accelerators
Francis Osswald, M. Chahbaoui, X. Liang (2026)

EPJ Research Infrastructures

Machine Learning Optimization of E-Beam Transport for a Superradiant FEL
Amir Weinberg, Leon Feigin, Ariel Nause, Avraham Gover (2026)
arXiv (Cornell University)

Automating Accelerator Tuning at GSI/FAIR
Sabrina Appel, Hendrik Alsmeier, M. Bajzek, Oliver Boine■Frankenheim, Lisa Dingeldein, Rolf Findeisen, Benjamin Halilovi■, Simon Hirlaender, Sebastian Hirt, Daniel Kallendorf, E. Kazantseva, O. Kazinova, Maximilian Kraeft, Eric Lenz, N. Madysa, A. Obertelli, Maik Pfefferkorn, S. Piétri, Christoph Reinwald, M. Schlaich, Stefan Sorge, H. Weick, F. Wienholtz (2026)
EPJ Research Infrastructures

Particle tracking with physics-informed deep learning methods
Matthias Remta, Anja Beck, Shanthalakshmi Kilambi, Thomas Zhang, Francesco Velotti (2026)
arXiv (Cornell University)

Contrastive Learning for Interpretable Anomaly Detection at Collider Experiments
Haoyi Jia, Sagar Addepalli, Julia Gonski (2026)
arXiv

Relative and absolute dosimetric commissioning of the ELIMAIA--ELIMED laser-driven proton beamline at 23.45 MeV
G. A. P. Cirrone, G. Petringa, A. Kurmanova, R. Catalano, A. Amato, A. Pappalardo, M. G. Guarrera, F. Schillaci, P. Blaha, F. Grepl, M. Tryus, A. Velyhan, L. Giuffrida, D. Margarone, G. Cuttone (2026)
arXiv

Bayesian Optimization of Molybdenum-99 Production by Laser Wakefield Acceleration Using Coupled PIC and Monte Carlo Simulations
Bruno Silveira Nunes, Nilson Dias Vieira Junior, Mirko Salomón Alva Sánchez, Alexandre Bonatto, Ricardo Elgul Samad (2026)
arXiv

Flow-based surrogate models for particle tracking
Matthias Remta, Yann Dutheil, Francesco Velotti (2026)
arXiv

On-Detector Machine Learning for Beam-Induced Background Rejection at a 10 TeV Muon Collider
Abadjiev, Daniel, Howard, Eliza, You, Tsz Ngong, Michaud, Ryan, Roberts, Benjamin Ryan, Rosser, Benjamin, Di Petrillo, Karri Folan, Badea, Anthony, Berry, Doug, Das, Arghya Ranjan, Dickinson, Jennet, Di Guglielmo, Giuseppe, Gupta, Harshul, Fahim, Farah, Gandrakota, Abhijith, Gray, Lindsey, Hirschauer, James, Jiang, David, Kuang, Shiqi, Lipton, Ron, Littmann, Mira, Liu, Miaoyuan, Manganelli, Nicholas, Maksimovic, Petar, Mills, Corrinne, Neubauer, Mark S., Nicholas, Aidan, Parpillon, Benjamin, Pearkes, Jannicke, Quinn, Adam, Shekar, Danush, Silvestre, Ricardo, Syal, Chinar, Swartz, Morris, Tran, Nhan, Trivedi, Amit, Ulmer, Keith, Wadud, Mohammad Abrar, Weiss, Benjamin (2026)
arXiv

Exploration of Real Time Inference for MI-RR Deblending on GPU/TPU Systems
Yu, Kellen (2026)
InspireHEP

Machine learning for slow-extraction uniformity at the Fermilab Delivery Ring
Prescott, Matthew J. (2026)
InspireHEP

Gaussian Process Regression and Bayesian Optimization for a 40-90 MeV Laser-Plasma Injector for the cSTART Storage Ring
David Squires, Elias Sailer, Joseph Natal, Alexander Saw, Nathan Ray, M. Fuchs, Thiemo

Schmelzer (2026)
KITopen

Machine Learning Based ROI Segmentation for Beam Imaging Diagnostics at Accelerators
Prachiti Sujit Chandratreya, Frank Mayet, Sergey Tomin, Jitendra Kumar (2026)
arXiv (Cornell University)

LinacNet: A Composable Particle-Cloud Surrogate Model for Linear Accelerators, and Its Path to Larger Machines
Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2026)
Research Square

Longitudinal Phase Space Tomography via Machine Learning at the Los Alamos Neutron Science Center
Christopher Leon, Alexander Scheinker, Petr M. Anisimov (2026)
Research Square

HELIX: a hybrid envelope-multiparticle linac code with differentiable space-charge optimization
Abhishek Pathak (2026)
arXiv

Compact and efficient linear accelerator phase space modeling via implicit neural representation learning
Veng Jean Heng, Serdar Charyyev, Yong Yang, Cynthia Chuang, Lei Xing, Lianli Liu (2026)
Physics in Medicine & Biology

Flexible Spectral-Normalized Neural Gaussian Process for Dynamic Aperture Prediction
Yousra El-Bachir, Frederik Van der Veken, Davide di Croce, Carlo Emilio Montanari, Massimo Giovannozzi, Ekaterina Krymova, Tatiana Pieloni (2026)
arXiv (Cornell University)

Likelihood-Based Unsupervised Anomaly Detection in CMS Dijet Events
Bhavishya Chebrolu, Hitesh Rasineni, Prajwal Aaryan Immadi (2026)
arXiv

Transverse beam profile reconstruction using a hybrid neural network framework for wall current monitor at LEHIPA
Monika, Jose V. Mathew, Rajesh Kumar (2026)
Journal of Instrumentation

Physics-Informed Drift Diagnosis for Laser-Plasma Accelerator Operations
Ou Labun, Calin Hojbota, Mara Klebonas, Mike Downer, Rafal Zgadzaj, Phil Franke, Lance Labun (2026)
arXiv

Steady Spills, Stronger Signals: Machine Learning for Slow Spill Analysis
Matthew Prescott (2026)
Unknown Venue

Strategy of AI Application and Current Activities at European Spallation Source Proton Linac
Ishkhan Gorgisyan, T.J. Shea, Mamad Eshraqi, Karin Rathsman, Rafael Montaño, Mehdi Mohammednezhad, Hinko Kocevar, Philipp Arnold, Håkan Danared, M.J. Ferreira, P. Pierini, Morten Rostrup Forup Jensen, John G. Weisend, Kent Gustafsson, Gábor Perák, Domen Cvenkel, Karlis Berkolds, Maciej Grzegrzó■ka, Ciprian Plostinar (2026)
EPJ Research Infrastructures

AI-driven Neutrino Beam Diagnostics for Next-Generation Neutrino Experiments
Noah Aney, U. Chicago (main), Maxwell Chen, Sudeshna Ganguly (2026)
Unknown Venue

Similarity Pairing with Energy Mover's Distance for Self-Supervised Pre-Training at the LHC
Ho Fung Tsoi, Dylan Rankin (2026)

arXiv

Self-Supervised Learning for Robust Resonance Mass Regression in Cascade Decays
Ho Fung Tsoi, Alex Yang, Luis Felipe Gutierrez Zagazeta, Shion Chen, Dylan Rankin (2026)
arXiv

Moderator Modeling for High Intensity Slow Positron Sources
Sophie Crisp, Ryland Goldman, Spencer Gessner (2026)
arXiv

Bayesian Optimization of The Relativistic Heavy Ion Collider Luminosity via s^* Control
Xiaofeng Gu, Benjamin Coe, Guillaume Robert-Demolaize, Takeshi Kanesue, Masahiro Okamura, William Fung, Yue Hao, Ji Qiang, Xiaoye S. Li, Yang Liu (2026)
arXiv

Design and Operation of a 3-Crystal Multi-Volume Reflection Array for Loss Reduction in CERN SPS Slow Extraction
F. M. Velotti, M. Bay, T. Calvet, M. Calviani, G. Daher, L. S. Esposito, M. A. Fraser, B. Goddard, M. Howes, V. Kain, E. Matheson, A. Perillo Marcone, R. Seidenbinder (2026)
arXiv

Machine Learning for Invisible Dark Boson Searches at the Electron-Ion Collider
Rojae Mighty, Ankush Reddy Kanuganti (2026)
arXiv

Pulse-shape discrimination with machine learning for CZT detectors at the DAΦNE beam test facility
Manti, Simone, Artibani, Francesco, Abbene, Leonardo, Bazzi, Massimiliano, Bettelli, Manuele, Borghi, Giacomo, Bosnar, Damir, Bragadireanu, Mario, Buttacavoli, Antonino, Carminati, Mario, Clozza, Alberto, Clozza, Francesco, De Paolis, Luca, Del Grande, Raffaele, Dulski, Kamil, Fiorini, Carlo, Friš■i■, Ivica, Gerardi, Gaetano, Guaraldo, Carlo, Iliescu, Mihai, Iwasaki, Masa, Khreptak, Alexander, Marton, Johan, Moskal, Pawel, Ohnishi, Hiroaki, Piscicchia, Kristian, Principato, Fabio, Scordo, Alessandro, Sgaramella, Francesco, Silarski, Micha■, Sirghi, Diana, Sirghi, Florin, Skurzok, Magdalena, Spallone, Antonio, Toho, Kairo, Vazquez Doce, Oton, Zappettini, Andrea, Zmeskal, Johann, Curceanu, Catalina (2026)
arXiv

Performance evaluation of beam breakup instability analysis of energy recovery linac using physics-inspired neural networks
Joshi, M.K., Sethi, S.P., Setiniyaz, S., Nuchsirikulaphong, N., Apsimon, R. (2026)
Phys.Rev.Accel.Beams

Adaptive Interaction Graphs for Particle Simulation
Aiden Zhou (2026)
arXiv

Machine Learning for Longitudinal Monitoring and Control of Ultra-Bright Electron Beams
Johan Lundquist (2026)
Lund University Publications (Lund University)

Implementation and deployment of an injection tuning tool using Bayesian optimization at the SuperKEKB accelerator
Anonymous (2026)
Physical Review Accelerators and Beams

LampAttention: Look-Ahead Mixed-Precision FlashAttention for Dedicated Accelerators
Stanislav Budzinskiy, Marian Gloser, Tolunay Yilmaz, Ying Hong Tham, Yuanyi Lin, Wenyi Fang, Fan Wu, Philipp Petersen (2026)
arXiv

Reveal normal form structure for nonlinear map in accelerator beam physics with symplectic neural network

Leo He, Yue Hao (2026)
arXiv

The Multi-Office Accelerator Team (MOAT) Seed project
Jean-Luc Vay, Thorsten Hellert, A. Huebl, Remi Lehé, Gianluca Martino, Chad Mitchell, Edoardo Zoni, Auralee L. Edelen, Daniel F. Ratner, Ryan Roussel, Zhe Zhang, Noa Steerenberg, Chris Tennant, Adam Carpenter, Jerone Samari, Chenran Xu, Hairong Shang, Marty Smith, Philippe Piot, Hayden Hoschouer, Nikita Kuklev, Jonathan D. Jarvis, Meifeng Lin, Wenge Fu, Weijian Lin, K. A. Brown, Nathan Urban, Alexander Zhukov, Kiersten Ruisard (2026)
Research Square

Anomaly Detection in Bunch Oscillations for Early Warning of Sudden Beam Loss at SuperKEKB
H. Haigh, P. Horak, G. Inguglia, C. Krause, G. Mitsuka, H. Nakayama, R. Nomaru, M. Tobiyama (2026)
arXiv

Learn the Dynamics, Enforce the Physics: Generative Latent Operators for High-Dimensional Beam Evolution
Haohan Zou, Alexander Scheinker, Yuanyuan Shi (2026)
arXiv

Multi-Objective Design of C-Shaped Targets for Energy--Divergence Control in Laser-Driven Proton Acceleration
Mohammad Rezaei-Pandari, Mahdi Shayganmanesh, Mohammad Hossein Mahdieh (2026)
arXiv

Orbit correction for a rapid cycling synchrotron using an adjacent-difference-constrained neural network
Zhong, Hao, Lu, Xiaohan, Bao, Wei, An, Yuwen, Li, Yong, Chen, Jianliang, Liu, Hanyang, Xu, Shouyan, Zhang, Yuliang, Huang, Mingyang, Jiao, Yi, Wang, Sheng (2026)
Phys.Rev.Accel.Beams

Automated end-station at the soft X-ray beamline featuring robotic air-to-ultrahigh-vacuum sample transfer, scanning probe spectroscopy, and machine learning analysis
Hiromori, Keita, Miyamoto, Masashige, Horio, Masafumi, Nakashima, Takeru, Ando, Yasunobu, Kotsugi, Masato, Matsuda, Iwao (2026)
arXiv

Physics-constrained graph closures and working-point conditioning for a fast surrogate of linac beam dynamics
Bowen Zhang, Dazhang Huang, Nengyuan Zhang, Duan Gu, Chao Feng (2026)
arXiv